# *Band Representation Based Machine Learning of Crystal*

## *Atom, Bond and Crystal Representations*

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# *Preview*

- Task:

  Machine Learning(ML) for crystal property predictions:

  band gap,  formation energy, metal/insulator classification, …

- Difficulty:
  - How to input a crystal structure as a feature vector?
  - Find a universal and interpretable representation of crystals

Atom coordinates?
Space group symmetry?
Local chemical environment?

# *Atom and bond feature in molecular graph*

**Table 1. Atom Features for the MG Representation**[a]

| feature | description |
|---|---|
| atom type | H, C, N, O, F (one-hot) |
| chirality | R or S (one-hot or null) |
| formal charge | integer electronic charge |
| ring sizes | for each ring size (3–8), the number of rings that include this atom |
| hybridization | $sp$, $sp^2$, or $sp^3$ (one-hot or null) |
| hydrogen bonding | whether this atom is a hydrogen bond donor and/or acceptor (binary values) |
| aromaticity | whether this atom is part of an aromatic system |

[a]Values are provided for each atom in the molecule.

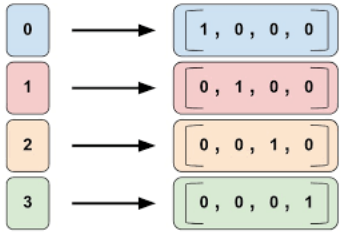
**Table 2. Atom Pair Features for the MG Representation**[a]

| feature | description |
|---|---|
| bond type | single, double, triple, or aromatic (one-hot or null) |
| graph distance | for each distance (1–7), whether the shortest path between the atoms in the pair is less than or equal to that number of bonds (binary values) |
| same ring | whether the atoms in the pair are in the same ring |
| spatial distance | the Euclidean distance between the two atoms |

[a]Values are provided for each pair of atoms in the molecule.

- Atom feature

  One-hot encoding for every category

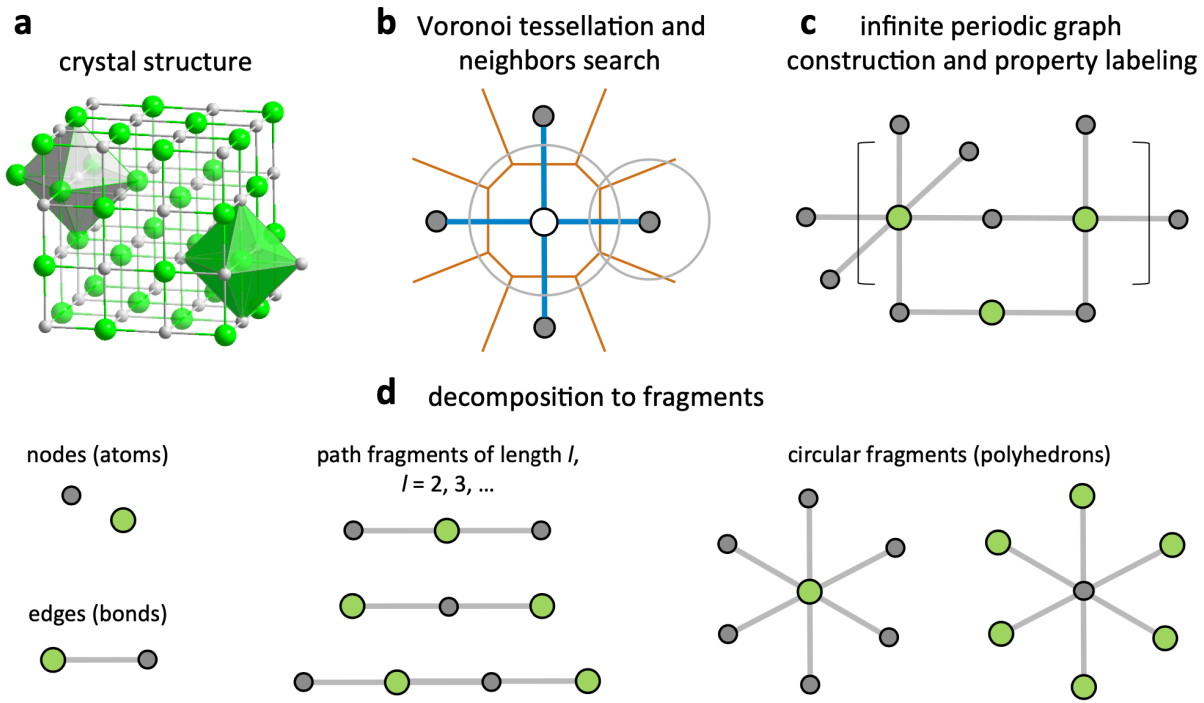



- Bond feature

  Bond type, graph distance, spatial distance, whether in the same ring

F. A. Faber *et al*, *J. Chem. Theory Comput.*, **13**, 5255 (2017)

# *Graph representation of crystals*




- Atom(element) feature

  Group and period number, atomic mass, electron affinity, … 2494 descriptors in total

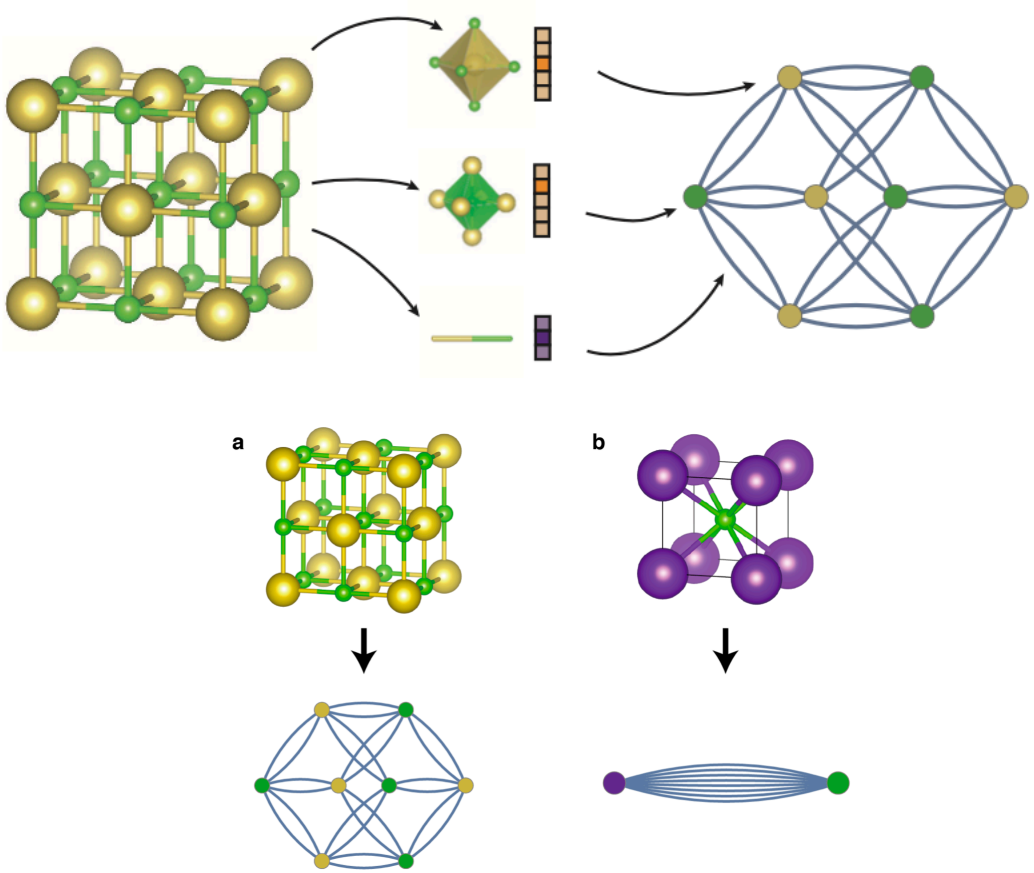
- Graph representation of crystals

  Galvez matrix:

$$\mathbf{M} = \mathbf{A} \cdot \mathbf{D} \quad \left(D_{ij} = 1/r_{i,j}^2\right)$$

O. Isayev *et al*, *Nat. Commun*. **8**. 15679 (2017)

# *Crystal Graph Convolutional Neural Networks(CGCNN)*




- Atom feature

  Group and period #, electronegativity, covalent radius, # of valence electrons, first ionization energy, electron affinity, s/p/d block, atomic volume

- Bond feature

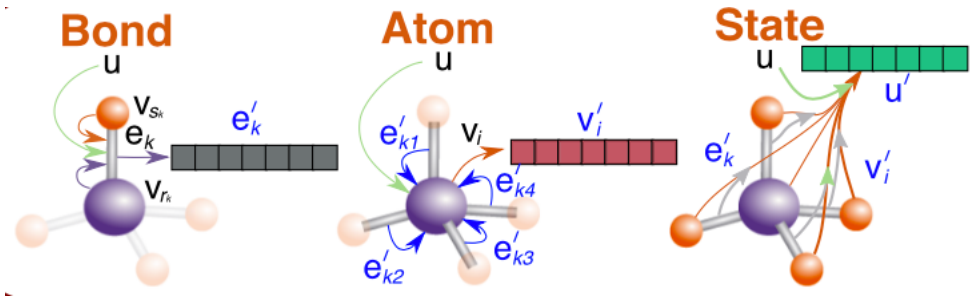
  Expand the distance in the basis of a series of Gaussians

- Crystal graph

  Topological, Crystalline symmetry

T. Xie and J. C. Grossman, *Phys. Rev. Lett*. **120**.145301 (2018)

# *MatErials Graph Network(MEGNet)*




$$\mathbf{v}_{s_k} \oplus \mathbf{v}_{r_k} \oplus \mathbf{e}_k \oplus \mathbf{u}$$

- Atom feature:
  - Molecule: atom type, chirality, hybridization, acceptor & donor…
  - Crystal: atomic number
- Bond feature

  expanded in Gaussian basis
- State feature(global)
  - e.g., temperature

Chi Chen, Weike Ye, Yunxing Zuo, Chen Zheng, and Shyue Ping Ong, *Chem. Mater*. 31, 3564 (2019)

## *Better representations?*

Can we make use of real space and reciprocal space symmetry information?

# Group Representation Theory

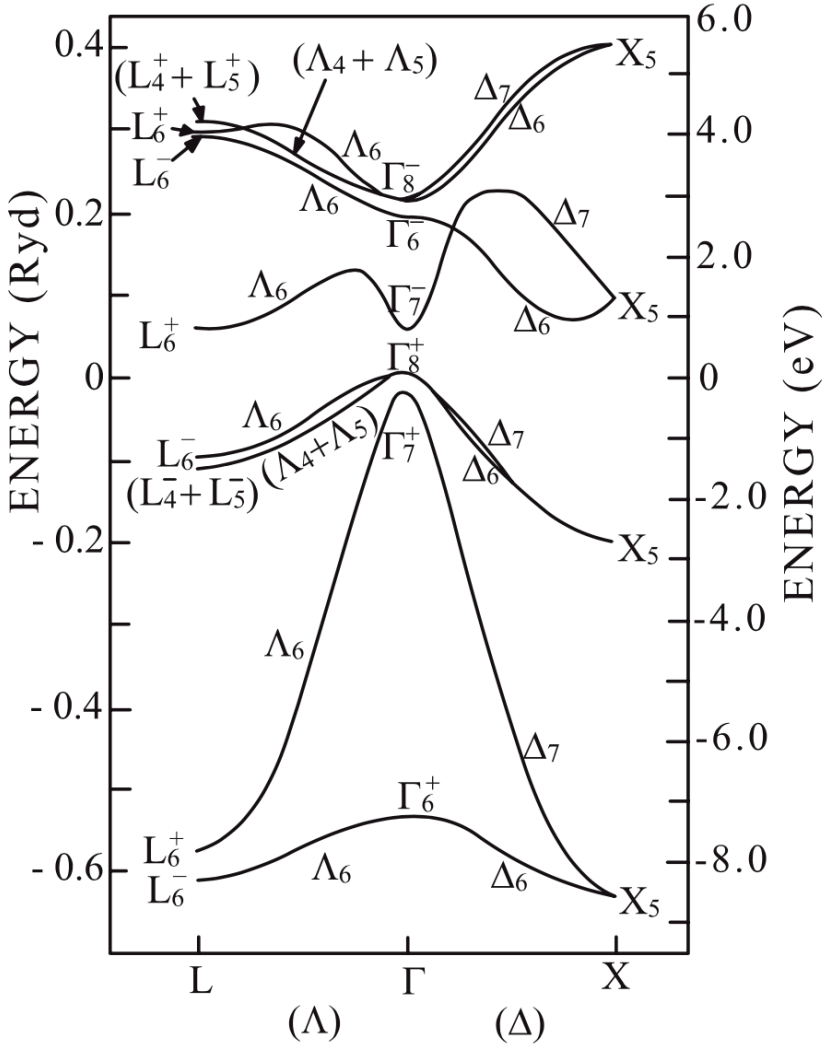



**Table 14.6.** Double group character table for the group $O$

| $O$ | $E$ | $\mathcal{R}$ | $3C_4^2 + 3\mathcal{R}C_4^2$ | $6C_4$ | $6\mathcal{R}C_4$ | $6C_2' + 6\mathcal{R}C_2''$ | $8C_3$ | $8\mathcal{R}C_3$ |
|---|---|---|---|---|---|---|---|---|
| $\Gamma_1$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| $\Gamma_2$ | 1 | 1 | 1 | $-1$ | $-1$ | $-1$ | 1 | 1 |
| $\Gamma_{12}$ | 2 | 2 | 2 | 0 | 0 | 0 | $-1$ | $-1$ |
| $\Gamma_{15'}$ | 3 | 3 | $-1$ | 1 | 1 | $-1$ | 0 | 0 |
| $\Gamma_{25'}$ | 3 | 3 | $-1$ | $-1$ | $-1$ | 1 | 0 | 0 |
| $\Gamma_6$ | 2 | $-2$ | 0 | $\sqrt{2}$ | $-\sqrt{2}$ | 0 | 1 | $-1$ |
| $\Gamma_7$ | 2 | $-2$ | 0 | $-\sqrt{2}$ | $\sqrt{2}$ | 0 | 1 | $-1$ |
| $\Gamma_8$ | 4 | $-4$ | 0 | 0 | 0 | 0 | $-1$ | 1 |

- Take home messages:
  - Every band is an irreducible representation(irrep)
  - Dimension of irrep = band degeneracy

# *Group Representation Theory*

- Irreps may become reducible (split) going from high symm to low symm

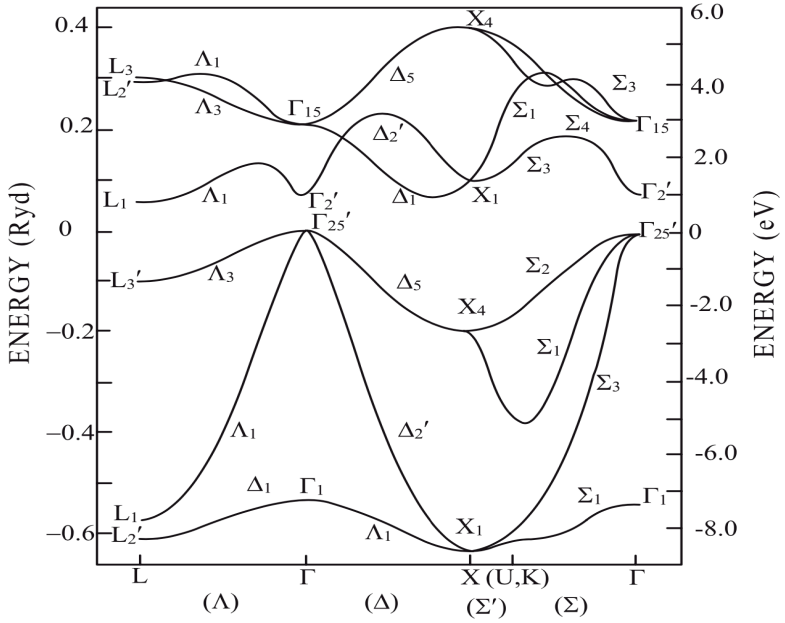
Crystal field splitting of atomic orbitals

| | $O$ | $E$ | $8C_3$ | $3C_2 = 3C_4^2$ | $6C_2'$ | $6C_4$ |
|---|---|---|---|---|---|---|
| $\Gamma_1$ | $A_1$ | 1 | 1 | 1 | 1 | 1 |
| $\Gamma_2$ | $A_2$ | 1 | 1 | 1 | $-1$ | $-1$ |
| $\Gamma_{12}$ | $E$ | 2 | $-1$ | 2 | 0 | 0 |
| $\Gamma_{15'}$ | $T_1$ | 3 | 0 | $-1$ | $-1$ | 1 |
| $\Gamma_{25'}$ | $T_2$ | 3 | 0 | $-1$ | 1 | $-1$ |
| $\Gamma_{\ell=0}$ | $A_1$ | 1 | 1 | 1 | 1 | 1 |
| $\Gamma_{\ell=1}$ | $T_1$ | 3 | 0 | $-1$ | 1 | $-1$ |
| $\Gamma_{\ell=2}$ | $E + T_2$ | 5 | $-1$ | 1 | 1 | $-1$ |
| $\Gamma_{\ell=3}$ | $A_2 + T_1 + T_2$ | 7 | 1 | $-1$ | $-1$ | $-1$ |
| $\Gamma_{\ell=4}$ | $A_1 + E + T_1 + T_2$ | 9 | 0 | 1 | 1 | 1 |
| $\Gamma_{\ell=5}$ | $E + 2T_1 + T_2$ | 11 | $-1$ | $-1$ | $-1$ | 1 |

Splitting along high symmetry lines

$$\Gamma_{25}' \to \Delta_{2'} + \Delta_5 \to X_3 + X_5\,,$$

$$\Gamma_{25}' \to \Lambda_1 + \Lambda_3 \to R_{15}\,,$$



# *Wyckoff positions*

**Definition 1.** *The set of symmetry operations, $g \in G$, that leave the site* $\mathbf{q}$ *fixed is called the* **stabilizer group** *or* **site-symmetry group** *of* $\mathbf{q}$*, and is denoted* $G_{\mathbf{q}} \equiv \{g | g\mathbf{q} = \mathbf{q}\} \subset G$.

- A site-symm group is always isomorphic to a crystallographic point group

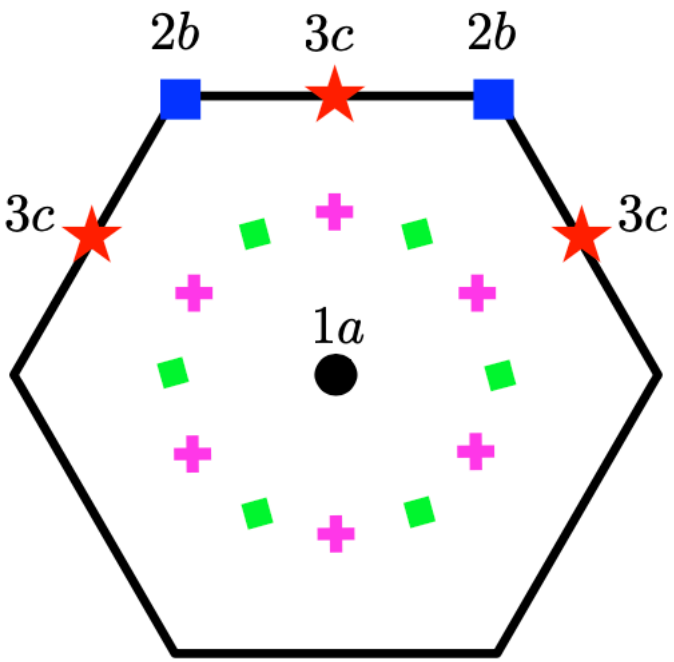
**Definition 2.** *Any two sites whose site-symmetry groups are conjugate are said to lie in the same* **Wyckoff position.** *Given a site in the Wyckoff position, the number of sites in its orbit that lie in a single unit cell defines the* **multiplicity** *of the position.*

$$G_{\mathbf{q}_\alpha} \equiv \{g_\alpha h g_\alpha^{-1} | h \in G_{\mathbf{q}}\} \qquad g_\alpha \mathbf{q} = \mathbf{q}_\alpha$$

B. Bradlyn et al, *Nature* **547**, 298 (2017)  J. Cano et al, *Phys. Rev. B*, **97**.035139 (2018)

# *Wyckoff positions*

- Example: graphene




$$G_{\boldsymbol{q}_a} \approx C_{6v}, \; G_{\boldsymbol{q}_b} \approx C_{3v}, \; G_{\boldsymbol{q}_c} \approx C_{2v}$$

B. Bradlyn et al, *Nature* **547**, 298 (2017)  J. Cano et al, *Phys. Rev. B*, **97**.035139 (2018)

# Band representation

**Definition 4.** *The* **band representation** $\rho_G$, *induced from the* $n_q$*-dimensional representation,* $\rho$*, of the site-symmetry group,* $G_{\mathbf{q}}$*, of a particular point,* $\mathbf{q}$*, whose orbit contains the sites* $\{\mathbf{q}_\alpha \equiv g_\alpha \mathbf{q}\}$ *in the unit cell, is defined by the action*

$$(\rho_G(h)a)_{i\alpha}(\mathbf{k},\mathbf{r}) = e^{-i(R\mathbf{k})\cdot\mathbf{t}_{\beta\alpha}} \times \sum_{i'=1}^{n_{\mathbf{q}}} \rho_{i'i}(g_\beta^{-1}\{E|-\mathbf{t}_{\beta\alpha}\}hg_\alpha)a_{i'\beta}(R\mathbf{k},\mathbf{r}), \tag{5}$$

*for each* $h = \{R|\mathbf{v}\} \in G$*, where for each choice of* $\alpha$ *the index* $\beta$ *is determined by the unique coset of* $G$ *in Eq (2) that contains* $hg_\alpha$*:*

$$hg_\alpha = \{E|\mathbf{t}_{\beta\alpha}\}g_\beta g \tag{6}$$

*for some* $g \in G_{\mathbf{q}}$*, coset representative* $g_\beta$*, and Bravais lattice vector* $\mathbf{t}_{\beta\alpha}$*.*

The choice of coset representatives, $g_\alpha$, must be kept fixed throughout the construction. The translation $\mathbf{t}_{\beta\alpha}$ is found as follows: by moving $g_\alpha$ to the right-hand-side of Eq (6), it is evident that $h\mathbf{q}_\alpha = \{E|\mathbf{t}_{\beta\alpha}\}g_\beta g g_\alpha^{-1}\mathbf{q}_\alpha = \{E|\mathbf{t}_{\beta\alpha}\}g_\beta g\mathbf{q} = \{E|\mathbf{t}_{\beta\alpha}\}g_\beta\mathbf{q} = \{E|\mathbf{t}_{\beta\alpha}\}\mathbf{q}_\beta$ (the second and fourth equalities follow from the definition of $\mathbf{q}_{\alpha,\beta}$ and the third equality follows from $g \in G_{\mathbf{q}}$), which yields

$$\mathbf{t}_{\beta\alpha} = h\mathbf{q}_\alpha - \mathbf{q}_\beta. \tag{7}$$

- For each $h = \{R|\mathbf{v}\} \in G$ ,

$$\rho_G^{\mathbf{k}}(h)_{j\beta,i\alpha} \equiv e^{-i(R\mathbf{k})\cdot\mathbf{t}_{\beta\alpha}}\rho_{ji}(g_\beta^{-1}\{E|-\mathbf{t}_{\beta\alpha}\}hg_\alpha)$$

The characters of $\rho_G \downarrow G_{\mathbf{k}}$ are given by, for $h \in G_{\mathbf{k}}$,

$$\boxed{\chi_G^{\mathbf{k}}(h) \equiv \sum_\alpha e^{-i(R\mathbf{k})\cdot\mathbf{t}_{\alpha\alpha}}\tilde{\chi}\left[\rho(g_\alpha^{-1}\{E|-\mathbf{t}_{\alpha\alpha}\}hg_\alpha)\right]}, \tag{9}$$

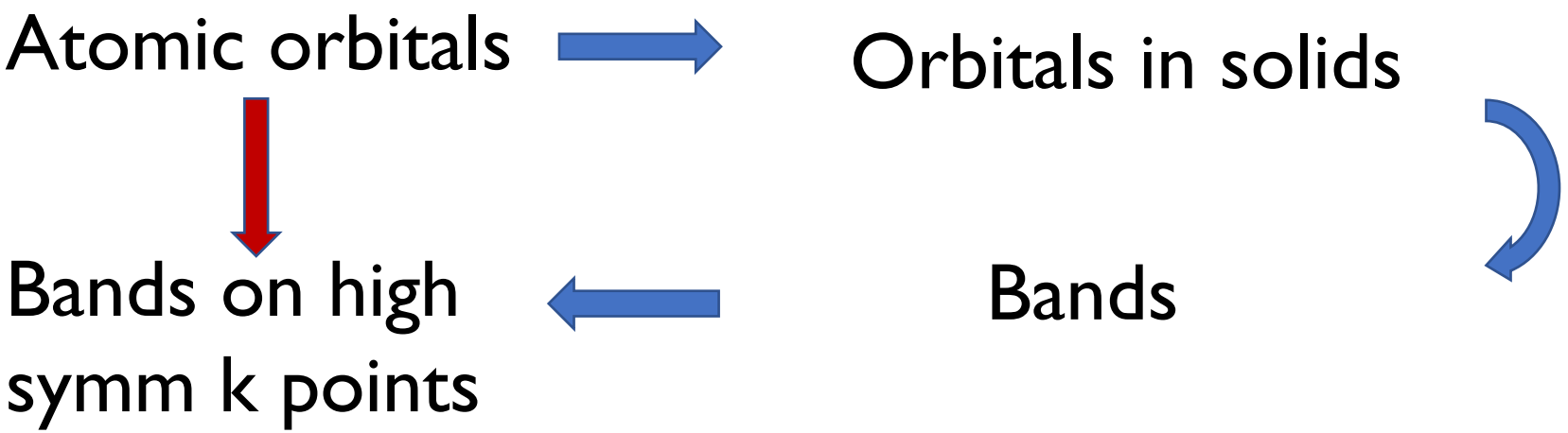
where

$$\tilde{\chi}[\rho(g)] = \begin{cases} \chi[\rho(g)] & \text{if } g \in G_{\mathbf{q}} \\ 0 & \text{if } g \notin G_{\mathbf{q}} \end{cases} \tag{10}$$

B. Bradlyn et al, *Nature* **547**, 298 (2017)  J. Cano et al, *Phys. Rev. B*, **97**.035139 (2018)

# *Band representation*

- Workflow

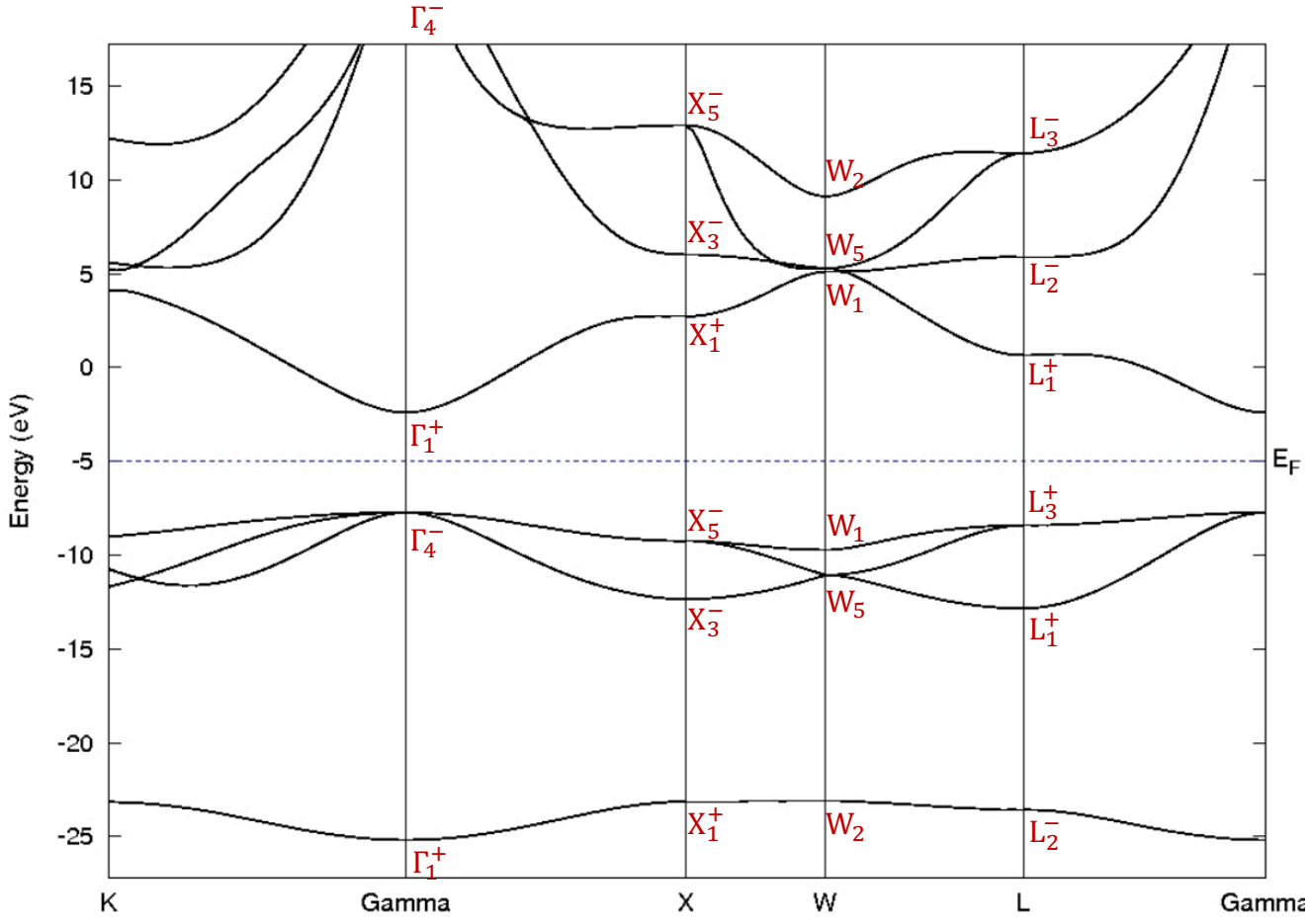



B. Bradlyn et al, *Nature* **547**, 298 (2017)  J. Cano et al, *Phys. Rev. B*, **97**.035139 (2018)

# *Band representation*

- Example: MgO

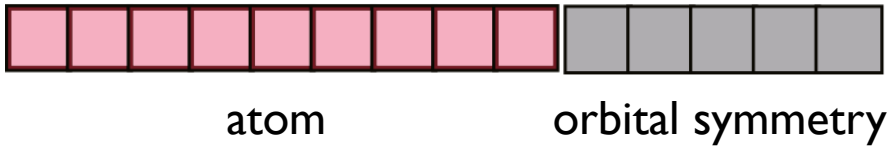
| Mg s | Mg p | O s | O p |
|---|---|---|---|
| 4a($m\bar{3}m$) | 4a($m\bar{3}m$) | 4b($m\bar{3}m$) | 4b($m\bar{3}m$) |
| $A_{1g} \uparrow G(1)$ | $T_{1u} \uparrow G(3)$ | $A_{1g} \uparrow G(1)$ | $T_{1u} \uparrow G(3)$ |
| Indecomposable | Indecomposable | Indecomposable | Indecomposable |
| $\Gamma_1^+(1)$ | $\Gamma_4^-(3)$ | $\Gamma_1^+(1)$ | $\Gamma_4^-(3)$ |
| $X_1^+(1)$ | $X_3^-(1) \oplus X_5^-(2)$ | $X_1^+(1)$ | $X_3^-(1) \oplus X_5^-(2)$ |
| $L_1^+(1)$ | $L_2^-(1) \oplus L_3^-(2)$ | $L_2^-(1)$ | $L_1^+(1) \oplus L_3^+(2)$ |
| $W_1(1)$ | $W_2(1) \oplus W_5(2)$ | $W_2(1)$ | $W_1(1) \oplus W_5(2)$ |

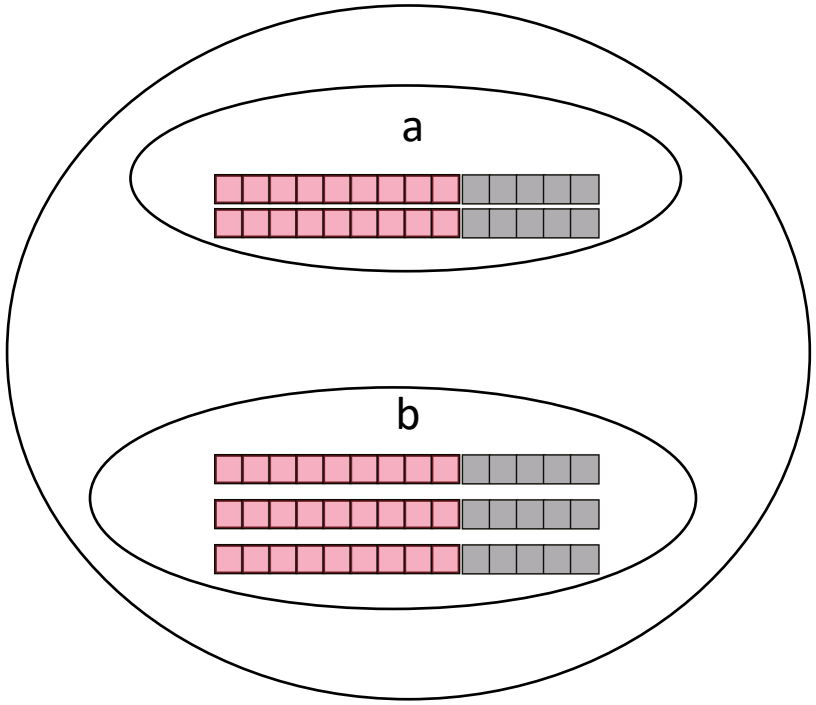



B. Bradlyn et al, *Nature* **547**, 298 (2017)  J. Cano et al, *Phys. Rev. B*, **97**.035139 (2018)

# *Feature vector*

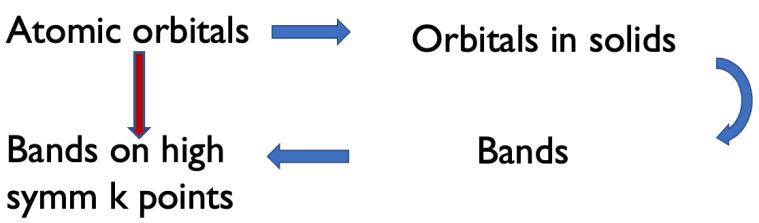
- Combine atom(element) feature and symmetry feature

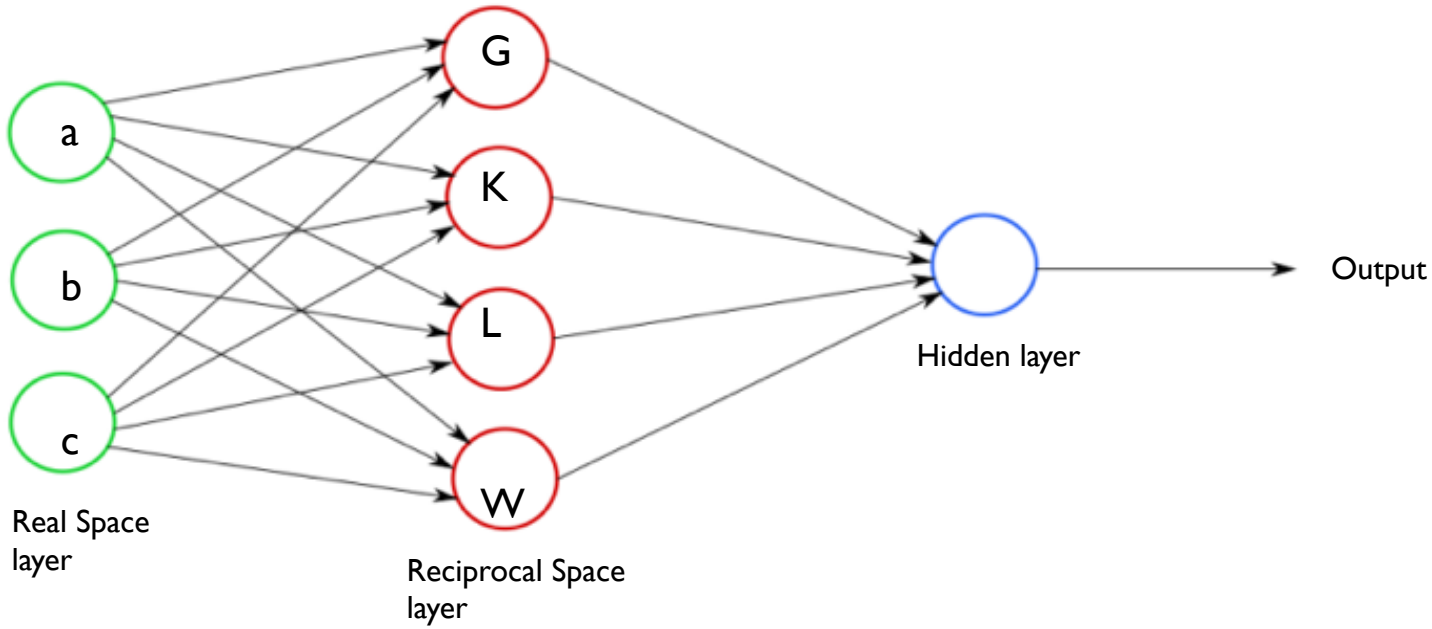


# *Neural Network*

- Possible network



***To be continue…***